THE N-1 PROPERTY DESCENDS FROM A LAURENT RING TO THE NOETHERIAN DOMAIN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Integral closure inside a normal Laurent ring. Used at the start of the counted proof (lines 45458-45461). Places the integral closure of the Laurent ring inside the Laurent ring of the fraction field.
- Finiteness of submodules of a finite module over a Noetherian ring. Used at the end of the counted proof (lines 45468-45470). Shows the integral closure of R is a finite R-module once it is contained in a finite one.

Lemma 0.1. *Let R be a Noetherian domain. If $R[z, z^{-1}]$ is N-1, then so is R.*

Proof. Let R' be the integral closure of R in its field of fractions K. Let S' be the integral closure of $R[z, z^{-1}]$ in its field of fractions. Clearly $R' \subset S'$. Since $K[z, z^{-1}]$ is a normal domain we see that $S' \subset K[z, z^{-1}]$. Suppose that $f_1, \ldots, f_n \in S'$ generate S' as $R[z, z^{-1}]$-module. Say $f_i = \sum a_{ij} z^j$ (finite sum), with $a_{ij} \in K$. For any $x \in R'$ we can write

$$x = \sum h_i f_i$$

with $h_i \in R[z, z^{-1}]$. Thus we see that R' is contained in the finite R-submodule $\sum R a_{ij} \subset K$. Since R is Noetherian we conclude that R' is a finite R-module. $\square$

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